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/*
* C program to accept N numbers sorted in ascending order
* and to search for a given number using Binary Search.
* Report success or failure.
*/

#include <stdio.h>

void main()
{
    int array[10];

    int i, j, num, temp, keynum;

    int low, mid, high;

    printf("Enter the value of num \n");

    scanf("%d", &num);

    printf("Enter the elements one by one \n");

    for (i = 0; i < num; i++)
    {
        scanf("%d", &array[i]);
    }

    printf("Input array elements \n");

    for (i = 0; i < num; i++)
    {
        printf("%d\n", array[i]);
    }

    /* Bubble sorting begins */

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for (i = 0; i < num; i++)
{
    for (j = 0; j < (num - i - 1); j++)
    {
        if (array[j] > array[j + 1])
        {
            temp = array[j];
            array[j] = array[j + 1];
            array[j + 1] = temp;
        }
    }
}

printf("Sorted array is...\n");

for (i = 0; i < num; i++)
{
    printf("%d\n", array[i]);
}

printf("Enter the element to be searched \n");

scanf("%d", &keynum);

/* Binary searching begins */

low = 1;

high = num;

do
{
    mid = (low + high) / 2;

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    if (keynum < array[mid])  
        high = mid - 1;  
    else if (keynum > array[mid])  
        low = mid + 1;  
} while (keynum != array[mid] && low <= high);  
if (keynum == array[mid])  
{  
    printf("SEARCH SUCCESSFUL \n");  
}  
else  
{  
    printf("SEARCH FAILED \n");  
}  
}
```